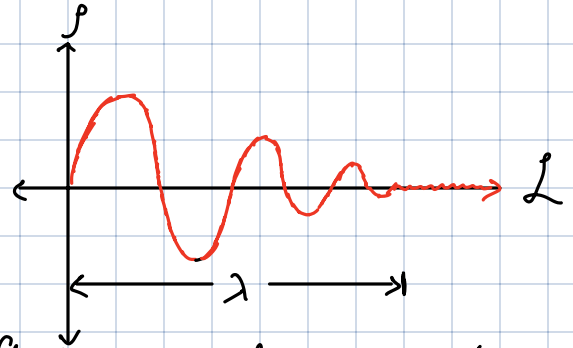
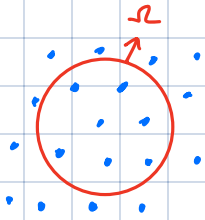


# ME 563: Intermediate Fluids

(Knudsen)  $Kn = \frac{\lambda}{l}$   $\lambda \equiv$  mean-free path

$$\rho = \frac{\sum \Omega (\text{molecules})}{\Omega}$$



If  $Kn < 0.01 \Leftrightarrow l > 100\lambda$ , then fluid can be regarded as fluid continuum

Eulerian:  $\tilde{x}$  is fixed in space  $\Rightarrow \partial_t f(\tilde{x}, t) = \frac{\partial f(\tilde{x}, t)}{\partial t}$  (temporal derivative)

Lagrangian:  $\tilde{x}$  is a moving reference  $\Rightarrow \partial_t f(\tilde{x}, t) = \frac{\partial f(\tilde{x}, t)}{\partial t} + \underbrace{\tilde{u} \cdot (\nabla f)}_{= (\tilde{u} \cdot \nabla) f}$  (convective derivative)

Boundary layer Equation:

$$u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} = \nu \frac{\partial^2 u}{\partial y^2}$$

$$\begin{cases} u=0 \text{ as } y \rightarrow 0 \\ u=u_\infty \text{ as } y \rightarrow \infty \end{cases}$$

$$x_i^* = \frac{x_i}{l} \quad (x_i' \equiv \text{characteristic length } l)$$

$$u_i^* = \frac{u_i}{u_\infty} \quad (u' \equiv \text{characteristic velocity } u_\infty)$$

$$\frac{\partial \psi}{\partial x_i} \rightarrow \frac{\partial \psi}{\partial (x_i^* l)} = \frac{1}{l} \frac{\partial \psi}{\partial x_i^*}$$

$$\frac{\partial^2 \psi}{\partial x_i^2} \rightarrow \frac{\partial^2 \psi}{\partial (x_i^* l)^2} = \frac{1}{l^2} \frac{\partial^2 \psi}{\partial x_i^{*2}}$$

$$u^* u' \left[ \frac{\partial (u^* u')}{\partial x} + \frac{\partial (u^* u')}{\partial y} \right] = \nu \left( \frac{1}{l^2} \right) \left[ \frac{\partial^2 (u^* u')}{\partial v^{*2}} \right]$$

$$u^* u_\infty \left[ \frac{u_\infty}{l} \frac{\partial u^*}{\partial x^*} + \frac{u_\infty}{l} \frac{\partial u^*}{\partial y^*} \right] = \nu \left( \frac{u_\infty}{l^2} \right) \frac{\partial^2 u^*}{\partial v^{*2}}$$

$$u^* \left( \frac{u_\infty^2}{l} \right) \left[ \frac{\partial u^*}{\partial x^*} + \frac{\partial u^*}{\partial y^*} \right] = \nu \left( \frac{u_\infty}{l^2} \right) \frac{\partial^2 u^*}{\partial v^{*2}}$$

$$\boxed{u^* u_\infty \left[ \frac{\partial u^*}{\partial x^*} + \frac{\partial u^*}{\partial y^*} \right] = \frac{\nu}{l} \frac{\partial^2 u^*}{\partial v^{*2}}}$$

where:  $Re = \frac{u_\infty l}{\nu}$

$$\text{let: } \begin{cases} U=0 & \text{at } y=0 \\ U=U_\infty & \text{at } y=\infty \end{cases}$$

$$\Rightarrow \begin{cases} U^*=0 & \text{as } y^* \rightarrow 0 \\ U^*=1 & \text{as } y^* \rightarrow \infty \end{cases}$$

$$\text{OR} \quad \begin{aligned} &U^* U_\infty = 0 \quad \text{at } y^* I = 0 \\ &U^* = 0 \quad \text{at } y^* = 0 \\ &\text{and} \\ &U^* U_\infty = U_\infty \quad \text{at } y^* I \rightarrow \infty \\ &U^* = 1 \quad \text{at } y^* \rightarrow \infty \end{aligned}$$



$$\Delta p = h(U, D, \Delta x, \rho, \mu, \epsilon)$$

$$Q = N - R \quad \text{Where: } \begin{cases} N \equiv \# \text{ of dimensional params.} \\ R \equiv \# \text{ of necessary dims.} \\ \text{(Rank of dim. matrix)} \end{cases}$$

|     | $\Delta p$ | $\rho$ | $U$ | $D$ | $\Delta x$ | $\epsilon$ | $\mu$ |
|-----|------------|--------|-----|-----|------------|------------|-------|
| $m$ | 1          | 1      | 0   | 0   | 0          | 0          | 1     |
| $L$ | -1         | -3     | 1   | 1   | 1          | 1          | -1    |
| $t$ | -2         | 0      | -1  | 0   | 0          | 0          | -1    |

$$\det(\underline{A}) = 0 \Rightarrow \text{Rank} = 3$$

$$Q = N - R = 4 \quad (\text{nondimensional groups})$$

→ Need to select  $R$  repeating parameters

→ Must span the required dimensional space; hence, the dimensional matrix for these  $R$  parameters is nonsingular.

→ Do not choose dependent variable (i.e.  $\Delta p$ ), say:  $\rho, U, D$

$$\pi_1 = \Delta p \rho^a U^b D^c \quad \text{where: } [\Delta p] = m^0 L^0 t^0$$

$$M: 1 + a + 0 + 0 = 0$$

$$L: -1 - 3a + b + c = 0$$

$$t: -2 + 0 - b + 0 = 0$$

$$\Rightarrow \begin{cases} a = -1 \\ c = 0 \\ b = -2 \end{cases}$$

$$\Rightarrow \boxed{\pi_1 = \Delta p \rho^{-1} U^{-2}}$$

$$\pi_2 = \mu \rho^a U^b D^c$$

$$M: 1 + a = 0$$

$$L: -1 - 3a + b + c = 0$$

$$t: -1 - b = 0$$

$$\Rightarrow \begin{cases} a = -1 \\ b = -1 \\ c = -1 \end{cases}$$

$$\Rightarrow \boxed{\pi_2 = \mu \rho^{-1} U^{-1} D^{-1}}$$

$$\pi_3 = \Delta x \rho^a U^b D^c$$

$$M: 0 + a = 0$$

$$L: 1 - 3a + b + c = 0$$

$$t: -b = 0$$

$$\Rightarrow \begin{cases} a = 0 \\ b = 0 \\ c = -1 \end{cases}$$

$$\Rightarrow \boxed{\pi_3 = \Delta x D^{-1}}$$

$$\pi_4 = \varepsilon \rho^a u^b D^c$$

$$M: a=0$$

$$L: 1 - 3a + b + c = 0$$

$$t: -b = 0$$

$$\Rightarrow \begin{cases} a=0 \\ b=0 \\ c=-1 \end{cases}$$

$$\boxed{\pi_4 = \varepsilon D^{-1}}$$

$$\pi_1 = g(\pi_2, \pi_3, \pi_4) \quad ; \quad \frac{\Delta p}{\rho u^2} = g\left(\frac{1}{Re_0}, \frac{\Delta x}{D}, \frac{\varepsilon}{D}\right)$$

$$\Delta p \propto \Delta x \Rightarrow \frac{\Delta p}{\rho u^2} = \frac{\Delta x}{D} f\left(\frac{1}{Re_0}, \frac{\varepsilon}{D}\right)$$